



Revisiting recommender systems: an investigative survey

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Abstract

This paper provides a thorough review of recommendation methods from academic literature, offering a taxonomy that classifies recommender systems (RSs) into categories like collaborative filtering, content-based systems, and hybrid systems. It examines the effectiveness and challenges of these systems, such as filter bubbles, the “cold start” issue, and the reliance on collaborative filtering and content-based approaches. We trace the development of RSs, emphasizing the role of machine learning and deep learning models in overcoming these challenges and delivering more accurate, personalized, and context-aware recommendations. We also highlight the increasing significance of ethical considerations, including fairness, transparency, and trust, in the design of RSs. The paper presents a structured literature review, discussing various aspects of RSs, such as collaborative filtering, personalized recommender systems, and strategies to improve system robustness. It also points out the limitations of the existing approaches and suggests promising research directions for the future. In summary, this paper offers a comprehensive analysis of RSs, focusing on their evolution, challenges, and potential future improvements, particularly in enhancing accuracy, diversity, and ethical practices in recommendations.

Keywords Recommender systems · Trust-aware · Fairness · Bias

1 Introduction

As the list of choices available to individuals has expanded, so too has their inclination to seek assistance when making decisions. Traditionally, they would seek counsel and suggestions from family and acquaintances. Presently, the ubiquity of internet-connected desktop computers, laptops, and mobile devices has revolutionized their decision-making

processes. Simultaneously, the substantial proliferation of data accessible via the World Wide Web has heightened the demand for online decision support systems and recommendation systems. Additionally, social networking platforms like Twitter, Facebook, and LinkedIn have emerged as valuable tools for guiding individuals in achieving their objectives in alignment with their preferences and interests. Recommender systems (RSs), as indicated by prior research [1–4], have been devised to furnish internet users with valuable recommendations. These recommendations play a pivotal role in facilitating decision-making by offering tailored options. Essentially, RSs serve as navigational aids, directing individuals toward more suitable choices when they encounter a deficiency in information or expertise. The effectiveness and widespread acceptance of RSs in the realms of social networks and e-commerce have served as a catalyst for their integration into diverse domains, encompassing healthcare, commerce, finance, and travel agencies [4–6].

The emergence of filter bubbles, where users are exclusively exposed to content aligning with their existing preferences and biases, represents a significant concern within recommendation systems. This limited exposure to diverse content can potentially curtail users’ experiences with novel and heterogeneous information. Furthermore, challenges

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related to the “cold start” scenario may arise when a new item or user lacks historical data for recommendation purposes, as exemplified by previous studies [7, 8]. Consequently, the system may struggle to discern the user’s preferences, resulting in suboptimal recommendations. The effectiveness of recommendation systems in capturing the intricate and evolving landscape of user preferences and behaviors is further impeded by their reliance on collaborative filtering and content-based techniques. Consequently, newer recommendation approaches, such as machine learning (ML) and reinforcement learning, have emerged to address these constraints and deliver more accurate and personalized recommendations [8–10]. In the last few years, considerable progress has been made in the field of recommender systems due to the combination of collaborative filtering (CF) and ML, particularly in the area of managing complex, dynamic, and large-scale datasets [11–13]. With these advancements, it is possible to provide recommendations that are more accurate, customized, and context-sensitive. Deep learning models are one of the most significant developments in CF [14, 15]. By utilizing neural networks to detect nonlinear interactions between users and items, neural collaborative filtering (NCF) goes beyond traditional CF models like matrix factorization [16]. The modeling of complicated patterns and feature representations required for understanding implicit feedback and latent user preferences is enabled by deep learning architectures. The precision reports predicts¹ that the worldwide recommendation engine market will grow significantly between 2023 and 2030, indicating the growing importance of recommendation engines in corporate strategy. In recent years, social computing and human computer interaction (HCI), which have long emphasized the human element of technology, have given greater attention to human-AI interaction [17, 18]. A private knowledge gap occurs when there are specific, often industry-specific, technologies or techniques that have not been publicly disclosed or documented in academic literature [19, 20]. As funders seek to disseminate key research outputs that will deliver value to society, these aspects are likely to be subject to increased consideration, debate, and focus. The majority of recommender system research focuses on algorithmic improvements, and may not perform well in real-world scenarios where highly-scored outcomes are not aligned with user-facing results. Consequently, a more detailed approach that considers human aspects in the context of practical recommender system applications is required [21, 22] (Figs. 1 and 2).

To address these challenges and considerations, researchers have recently devised recommendation systems that incorporate additional information, such as contextual cues

and side information. These innovative approaches aim to enhance the precision and relevance of recommendations. Consequently, the development of fair and transparent recommendation systems to mitigate potential adverse impacts has gained heightened importance. In this context, the significance of exploring and crafting novel recommendation systems that offer more precise and diverse suggestions, while also addressing ethical concerns, cannot be overstated. As evidenced by prior research [23–25], the rapid proliferation of recommendation systems underscores the necessity of establishing and defining standard strategies in a clear and understandable framework. The contribution of the current study can be concisely outlined as follows:

1. The primary scholarly contribution lies in elucidating and succinctly encapsulating the classifications and attributes of three principal categories of recommendation systems, accompanied by their respective definitions.
2. The present study undertakes the organization and contextualization of previous research endeavors within the realm of recommender systems (RSs) by employing a systematic taxonomy. As depicted in Fig. 1, the visual representation illustrates diverse categories of RSs that warrant in-depth exploration during the forthcoming literature review. Specifically, the diagram delineates the following three principal categories of RSs: collaborative filtering, personalised recommender systems, and content-based recommender systems. These primary classifications further encompass sub-categories, which will be examined in greater detail in subsequent sections.
3. In our review of the relevant literature, we not only present our perspectives but also delineate the potential drawbacks associated with the discussed works. Furthermore, we put forth several promising avenues for future research in this specific domain.

The subsequent section provides a structured overview of the literature review. Collaborative filtering (CF) methodologies are comprehensively deliberated within Sect. 4. Section 5 offers an in-depth analysis of personalized techniques employed in recommender systems (RSs). The exploration of content-based RSs is contained within Sect. 6, and strategies pertaining to the robustness of RSs are introduced in Sect. 7. In conclusion, Sect. 10 offers a comprehensive summary and overarching insights.

2 Literature review methodology

In this section, our approach to collecting relevant papers for our survey is presented and then discuss how the proposed survey builds on existing research. A mixed-method, semi-systematic approach is used to collect relevant

¹ Precision Reports. Worldwide Market Research Report, and Industry Analysis, accessed on August 2024, <https://www.precisionreports.co/>

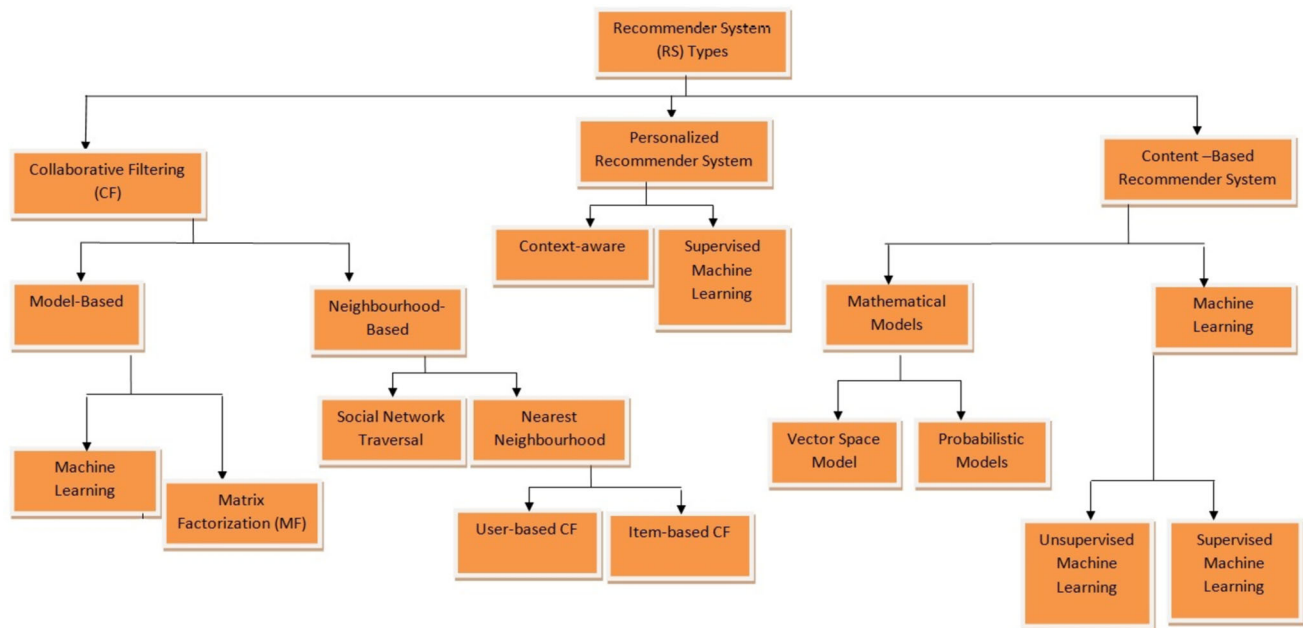
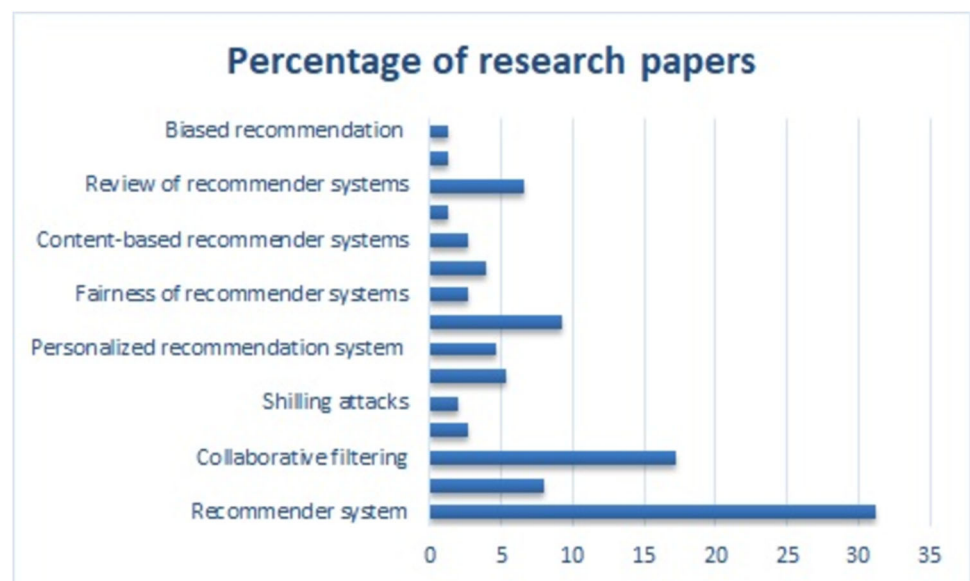


Fig. 1 Main types of recommender systems

Fig. 2 Percentage of research papers based on search keywords



research papers. The initial step involved querying the research papers publishing websites and libraries such as Springer, science direct and IEEE explore. To ensure that papers that might not match our search terms were included, we used a snowball technique leveraged the expertise of researchers to identify additional relevant papers from specialized sources. Given the evolving nature of this field and the lack of standardized terminology, we used a broad set of search terms to capture a wide range of relevant studies as depicted in 2. Terms included: recommender

system, trustworthy, collaborative filtering, e-commerce, shilling attacks, hybrid recommender system, personalized recommendation system, context-aware recommender system, fairness of recommender systems, deep learning for recommender systems, content-based recommender systems, recommender systems using generative models, review of recommender systems, ranking recommendations, and biased recommendation. The retrieved results are filtered such that with these search terms should appear in

the title or keywords of the papers while the papers that were merely preprints on arXiv.org are excluded.

Research Questions:

1. How has the theory of RS algorithms evolved over time?
2. How can theoretical developments in RS be translated into useful applications?

Search Query: (“recommender systems” OR “recommendation systems” OR “RS” OR “RecSys”) AND (“Latent Semantic Analysis” OR “LSA” OR “semantic analysis” OR “latent semantic models”) AND (“Rule-Based models” OR “rule-based recommendations” OR “expert systems” OR “rule-based algorithms” OR “knowledge-based systems”) AND (“matrix factorization” OR “deep learning” OR “neural collaborative filtering” OR “graph neural networks” OR “Large Language Models” OR “LLMs” OR “enhanced sequential recommendation” OR “sequential models” OR “collaborative filtering” OR “content-based filtering” OR “hybrid recommender systems”) AND (“precision” OR “recall” OR “F1 score” OR “RMSE” OR “MAE” OR “hit rate” OR “novelty” OR “diversity” OR “serendipity” OR “user satisfaction” OR “performance evaluation” OR “metrics for recommendation systems”) AND (“e-commerce” OR “media streaming” OR “social media” OR “education” OR “healthcare” OR “tourism” OR “personalized news” OR “job recommendation” OR “online retail” OR “content discovery” OR “medical decision-making” OR “social decision-making” OR “e-business decision-making”) AND (“algorithm evolution” OR “theoretical advancements” OR “practical implementation” OR “real-world applications” OR “effectiveness measurement” OR “optimization techniques” OR “scalability” OR “adaptability” OR “bias mitigation” OR “serendipity enhancement” OR “cold start problem” OR “data sparsity” OR “enhancing decision-making” OR “adaptive algorithms” OR “real-time recommendations” OR “contextual recommendations” OR “user behavior modeling”)

3 Previous related surveys

The purpose of this section is to highlight the differences between the current survey and earlier surveys. Earlier studies of recommender systems typically focused only on a single perspective domain, such as fairness [26]. In [27], the authors examined different design options for personality aware of RSs, comparing the personality modeling approaches, the recommendation techniques, as well as the datasets that have been used. A comprehensive survey [28] discussed the developments of RSs using generative models. It covered interaction-based generative models, the

use of large language models (LLM) and text data for natural language recommendation. It investigate the advantages and disadvantages of GAN models for RSs development process. Moreover, this study [29] explores the reasons behind popularity bias and reviews methods for detecting, quantifying, and mitigating it. Recently, [30] focuses on trustworthiness of RSs, through a comprehensive analysis of different aspects that can be used to characterize the reliability of assessment systems while this survey widely examines recommender systems from multiple perspectives/domains. It discusses and addresses the challenges and new techniques for each domain. These domains include collaborative filtering RSs, personalized RSs, content-based RSs, and robustness of RSs.

4 Collaborative filtering recommender systems (CF-RSs)

Collaborative filtering recommendation systems (CF-RSs) represent the most widely recognized recommendation systems employed on the internet. These systems function by sifting through an extensive array of item variations. In the case of CF-RSs, the filtration process is carried out collaboratively, harnessing the power of analogous user preferences. The fundamental premise that underlies CF-RSs is that if two users have exhibited similar preferences in the past, it is likely that their future desires will align as well. For instance, if both the first and second users have previously made analogous item selections, and the first user subsequently acquires a new product, such as a smartphone, which the second user has not yet encountered or acquired, the role of CF is to propose this previously undiscovered item to the second user. CF techniques make use of methods founded on the correlation between items or users. The latent semantic analysis (LSA) technique is used in information retrieval and natural language processing to analyze the relationships between a collection of documents and the terms they contain [31, 32]. Singular value decomposition (SVD) is used to find patterns and structures in the data and to capture underlying semantic links between words and documents [33]. In this way, it will be easier for the student to comprehend the meaning and context of terms that go beyond simple usage. The rule-based model is a class of recommender system that generates recommendations based on rules and heuristics that have been developed in advance [34, 35]. Domain experts are frequently responsible for developing these rules, which specify how suggestions should be generated based on specific conditions or criteria. Due to their explicit logic and rules for decision-making, rule-based models are usually transparent and easy to comprehend. These CF methods can be categorized into distinct groups:

- Memory-based techniques, often referred to as neighbourhood-based techniques, form a class of recommendation systems wherein the recommendations hinge upon the proximity of a user or item of interest to its neighboring counterparts. These proximities, which are calculated based on the similarity between the active user or the selected item and other users or latent pieces of information, are instrumental in the formation of these neighborhoods.
- Model-based methodologies, data mining, and machine learning techniques are employed in the construction of these models. These algorithms are instrumental in the development of predictive models tasked with generating recommendations. Illustrative instances of model-based methodologies encompass the random forest (RF), Bayesian models, latent semantic analysis, and rule-based models.

One fundamental differentiation between model-based and memory-based approaches pertains to the inclusion of a training phase within model-based techniques. Model-based methods entail a substantial investment of time in constructing a predictive model, which is subsequently employed for the purpose of forecasting and recommending information items to users of recommendation systems (RSs). Conversely, memory-based approaches do not involve a training stage and are predominantly dedicated to the task of identifying and suggesting information items. Consequently, the model-based approach proves to be better suited for real-time RSs that operate under resource constraints. Further elaboration on these two approaches is presented in the subsequent subsections.

4.1 Neighbourhood-based (memory-based) recommendation systems

Memory-based methodologies in the realm of recommender systems (RSs) are often referred to as neighbourhood-based approaches. These techniques involve the computation of similarity metrics among items or individuals within the RS, which are subsequently harnessed to recommend unseen items or products to users, predicated on proximity within the neighborhood, as established by these RSs.

4.1.1 Nearest neighbourhood

4.1.2 User-based CF

In the context of this particular recommendation system, the method for suggesting an item to an engaged user, denoted as “A,” relies on an approach that considers the evaluations or votes cast by other users on the items, as

discussed in Breese et al.’s work [36]. Let us suppose that $v_{k,i}$ represents the vote or evaluation assigned by user k to item i , and I_k denotes the set of items for which user k has provided evaluations. The computation of the average vote given by user k can be expressed as follows:

$$\bar{v}_k = \frac{1}{|I_k|} \sum_{i \in I_k} v_{k,i} \quad (1)$$

The predictive voting value, denoted as $P_{A,i}$, which guides the recommendation of an item i to an engaged user A, is derived from a weighted summation of preferences expressed by other users. The computation of this value can be formulated as follows:

$$P_{A,i} = \bar{v}_k + C \sum_{k=1}^n w(A, k)(v_{k,i} - \bar{v}_k) \quad (2)$$

A normalizing factor denoted as $C \sum_{k=1}^n w(A, k) = 1$ is employed to standardize the cumulative weight sum to unity. In this context, $w(A, k)$ signifies the degree of similarity or correlation weight characterizing the association between an active user, denoted as “A,” and another user indexed as “k.” There exist various methods to assess the correlations and relationships between the active user and other users. Notable examples encompass metrics like the Cosine and Pearson similarities. Further details on these similarity and correlation functions can be found in prior works, such as [36] and [37]. Additionally, for the determination of $w(A, k)$, supplementary extensions, including inverse user frequency and case amplification, as elucidated in [36], can be considered.

4.1.3 Item-based CF

Collaborative recommendation systems (RSs) employ a methodology wherein item recommendations for users are derived through an assessment of the similarity or correlation between items that the user has interacted with and those that have not yet been observed [38, 39]. This assessment relies on the user ratings provided for previously selected items in comparison to the new, unseen items. Let’s consider an active user, denoted as A, who has rated a selected item i and possesses a user rating vector represented as $\vec{i}$. Concurrently, there exists a user rating vector, denoted as $\vec{j}$, for a prospective item j that may be recommended to active user A. Consequently, the probability of item j being recommended to user A utilizing either the Cosine similarity or Pearson correlation, denoted as sim_{ij} or $corr_{ij}$, respectively, can be computed as follows:

$$sim_{ij} = cosine(\vec{i}, \vec{j}) = \frac{\vec{i} \cdot \vec{j}}{\|\vec{i}\| \|\vec{j}\|} \quad (3)$$

OR

$$corr_{ij} = \frac{\sum_{u \in U_{ij}} (r_{u,i} - \bar{r}_i)(r_{u,j} - \bar{r}_j)}{\sqrt{\sum_{u \in U_i} (r_{u,i} - \bar{r}_i)^2 \cdot \sum_{u \in U_j} (r_{u,j} - \bar{r}_j)^2}} \quad (4)$$

In the context of this study, the variable U_{ij} represents the count of users who have cast votes for both items i and j , while U_i and U_j denote the respective counts of users who have voted for item i and item j . Furthermore, the variables $r_{u,i}$ and $r_{u,j}$ correspond to the user ratings or votes attributed to items i and j . These ratings are represented by the vectors $\vec{i}$ and $\vec{j}$. Additionally, the symbol $\bar{r}$ signifies the mean rating value associated with an item.

4.1.4 Social network traversal (SNT)

In this methodology, participants within a social network are encouraged to contribute insights and assessments regarding information pieces or products to active network members, as indicated by [40]. This process encompasses three distinct categories of social network trust (SNT) techniques. The initial category is characterized by the trust weighted prediction method [37], followed by the Bayesian inference-based prediction method as the second category [41], and the random walk-based method as the third category [42]. It is essential to note that the SNT technique fundamentally relies on trust-aware methodologies, which will be comprehensively expounded upon in the subsequent Sect. 7.2.

4.2 Model-based recommendation systems

Model-based recommender systems (RSs) employ machine learning methodologies to construct predictive models derived from historical datasets. Subsequently, these models are harnessed to furnish personalized recommendations to end-users.

4.2.1 Machine learning techniques

Machine learning (ML) techniques are typically categorized into well-established paradigms, including supervised, unsupervised, and semi-supervised learning. Moreover, contemporary advancements in the field encompass state-of-the-art approaches, such as deep learning algorithms and transfer learning. A more comprehensive discussion of these paradigms and innovations is presented in Sect. 6.2 for detailed exploration.

4.2.2 Clustering-based CF

Clustering, within the context of collaborative filtering (CF), serves the purpose of organizing similar entities together based on user feedback's significance. Nevertheless, the clustering strategy employed for CF primarily hinges on grouping users according to their selection behavior within the recommendation system (RS). In this approach, users are characterized as vectors of rating features, where each feature corresponds to an element of information, a product, or a film, contingent upon the specific RS type under consideration. Subsequently, as elaborated in Sect. 4.1.1, similarity metrics are applied to amalgamate individuals who exhibit akin preferences. Ultimately, these user groups or clusters are leveraged to furnish recommendations of information or products to an individual based on the preferences of other users within the same cluster. Among clustering techniques, K-means stands out as one of the most prominent and widely utilized methods in this category.

4.2.3 Matrix factorisation (MF) method

The initial approach employed for creating the collaborative filtering with regularization strategy (CF-RS) centers on matrix factorization (MF), which shares similarities with the vector space model (VSM). In this approach, each user is represented by a vector comprising their ratings for the items within the recommendation system (RS), while each RS item is represented by a vector containing user ratings. The method proceeds by performing matrix factorization, which entails reducing the dimensions to a lower size through the decomposition of the user-items matrix. This is accomplished by formulating a linear equation that describes the relationship between the mean user ratings and user bias and item bias values. These equations are used to derive the anticipated ratings for the items within the RS. This process bears a resemblance to the principles of linear regression. The MF methodology can be further categorized into two primary approaches: 1) the similarity-based approach [43–45] and 2) the probabilistic MF method [46, 47].

The initial approach introduces a matrix factorization (MF) equation that is grounded in the probabilistic modeling of user ratings, user bias, and item rating bias. Conversely, the second approach presents an MF formulation that leverages user-item similarity within the context of a singular value decomposition (SVD) applied to the user-item rating matrix. It is noteworthy that the application of SVD for matrix factorization, as exemplified in the widely recognized Surprise Python Package [48], stands as a prominent and extensively employed MF methodology.

5 Personalised recommender systems (PRS)

5.1 Introduction to PRS

Recommender systems within personalized recommendation systems (PRS) construct their recommendation models by taking into consideration individual user attributes and pertinent user feedback. In the realm of recommender systems (RS), the journey commences with the formulation of recommendation models founded on collaborative filtering (CF) relevance feedback, which serves to address the issue of user cold start, primarily when users initially engage with the system and are unable to provide a sufficient volume of pertinent feedback labels. In this scenario, the machine learning approach discussed in Sect. 6.2.2 undergoes a training phase that relies on CF labels until a satisfactory quantity of personalized relevance labels becomes available. Recent trends in recommender systems and search engines have witnessed a growing adoption of machine learning techniques. Nevertheless, it remains uncertain whether these systems opt to utilize customized CF labels to exclusively tailor recommendations according to individual user preferences. Consequently, a noticeable gap exists within the literature concerning this innovative procedure.

5.2 Context-aware personalised recommender systems

In the realm of recommender systems (RSs), adaptive systems play a crucial role by delivering product recommendations to users that align with their prevailing environmental context. To illustrate, let us consider User A, who typically enjoys ice cream; however, if User A is presently afflicted with a cold, suggesting ice cream would not be appropriate. Achieving a high degree of personalization and relevance in recommendations necessitates the incorporation of various contextual factors, including but not limited to the user's location, the current time, the device being used, the user's social context, and a blend of different recommendation techniques [49–52]. Context-aware recommender systems, which incorporate contextual data (such as time, location, or social context), as well as user-item interaction data, are particularly prone to the cold start problem [53, 54]. When new users or things do not have adequate contextual information, it is significantly more difficult to make relevant recommendations. It is possible for this to occur under three different circumstances [55, 56]:

- **Problem with New Users:** There is no prior information on a new user's preferences or behavior. It will be difficult for a context-aware system to customize

recommendations based upon the context of a new user who has no interaction history in different contexts (e.g., different times of day, location, or device).

- **Problem with New Item:** The system may lack sufficient contextual interactions for a new item in order to understand how it performs under different circumstances, limiting its ability to recommend the item effectively to users.
- **New System Issue:** It is difficult to determine how context impacts user preferences and behaviors when a new context-aware system is first launched because it starts with minimal contextual data, making it difficult to determine how different contexts influence user preferences and behaviors.

5.3 Applications and examples of PRS

There exist diverse scenarios wherein RSs dynamically adapt their recommendations in response to user-specific requirements, such as during pregnancy or in response to emerging market trends, among other scenarios. The development of context-aware personalized recommender systems can take multiple approaches, depending on the specific objectives and constraints of the application. To provide context-aware recommendations tailored to a user's current situation, machine learning algorithms are employed to model the intricate interplay between contextual variables and user preferences, as evidenced in previous studies (e.g., [50, 51]). For instance, these algorithms can discern, based on factors such as time of day and prevailing weather conditions, the likelihood of a user favoring particular food choices. In parallel, rule-based systems (as documented in [52, 57]) are utilized to encapsulate domain-specific knowledge regarding the influence of contextual variables on user preferences. For example, within the domain of music streaming services, such systems might incorporate rules dictating that users tend to gravitate toward lively music genres during exercise routines and opt for mellower, soothing melodies before bedtime. Further comprehensive insights into context-aware recommender systems can be found in a seminal work by [49]. When context-aware recommender systems are combined with machine learning techniques, the cold start issue can be significantly reduced [55, 56]. The use of these techniques not only enhances the initial suggestions but also allows the system to quickly adjust to new data as it becomes available, ultimately improving the experience of the user through a more effective data collection process [53, 56]. A machine learning model was used by the authors in [58] to infer user preferences in the current domain based on data from several domains (such as social media preferences and search history). A machine learning

model can also incorporate contextual information into item embeddings [59]. In these embeddings, items are presented in a three-dimensional space that illustrates the relationships between objects, users, and environments. The context-aware matrix factorization method by Afzal et al. [60] was developed to extend classical matrix factorization by adding contextual information as an additional dimension. The method enables the system to predict user-item interactions despite sparse data. To accomplish this, the interaction is represented as a tensor containing user, item, and context information. An innovative method for feature engineering has been proposed in works such as [61–63]. To resolve this issue, Mao et al. [62] propose context-aware robust fine-tuning (CAR-FT). CAR-FT regularizes the model during fine-tuning in order to capture context information. A zero shot prompt weight is applied to the image in order to obtain the contextual distribution of the image. An integrated model for generating context-aware relation extraction is presented by Chen et al. [64]. Using the proposed methodology, a pre-trained prompt generator module creates relationship triggers based on context and embeds them into prompt tokens. Pretrained language models (PLMs) are enhanced by an adaptive pretraining module in the domain, and the model is optimized by a joint contrastive loss in order to prevent irrelevant content from being generated by the PLMs. The term multi-modal data refer to information that is represented in a variety of ways, such as through text, images, audio, or video. Multi-modal data can be processed and learned using machine learning models, including neural networks [65, 66]. The combination of product descriptions (text), images (visual features), and user-generated content (e.g., reviews, social media posts) allows machine learning models to predict item popularity and relevance across different contexts. The enriched item representation helps to overcome the cold start problem associated with new items [12, 67]. Visual data can be processed by convolutional neural networks (CNNs), while sequential text data can be processed by recurrent neural networks (RNNs) or transformer models [68–70]. By combining these models into a multimodal framework, different types of data can be processed at the same time, allowing complex interactions between the different types of data to be learned. In a graph structure, GNNs can be used to model relationships between users, items, and contextual information [68]. In this way, the system is capable of making inferences based on the fusion of data from various sources, even in the absence of direct connections (e.g., a sparse number of interactions between users and items). Context-aware strategies that modify the recommendations according to the particular environment in which the user operates can improve this transfer learning approach [47, 70]. Combining data fusion with advanced machine learning models

provides a robust solution to the cold start problem in recommender systems. The use of ML models can generate accurate and customized recommendations even with limited direct interaction data by integrating and analyzing diverse data sources such as user behavior, item metadata, and contextual information. In addition to reducing the initial challenges associated with cold start scenarios, this approach improves the overall effectiveness and adaptability of the recommender system across a variety of application domains.

6 Content-based recommender systems (CB-RSs)

The primary differentiating factor between content-based recommendation systems (CB-RSs) and collaborative filtering recommendation systems (CF-RSs) lies in their approach to providing recommendations. CB-RSs primarily rely on item and user characteristics to make predictions, while CF-RSs base their recommendations solely on consumer rating data associated with products. It is worth noting that CB-RSs are a subset of the broader field of information retrieval (IR), as both CB-RSs and IR are primarily concerned with the analysis of content data and the delivery of item information to consumers. Furthermore, recommendation systems (RSs) extend their reach beyond textual online content by incorporating additional data sources, such as the Semantic Web and Linked Data, as discussed by [71].

6.1 Mathematical models (MM)

The MM model delineates the structural framework by which a recommender system arranges, catalogues, and proffers information. Within the purview of MM models, two predominant categories emerge, namely, vector space models (VSM) and probabilistic models, as documented by [72]. These approaches adhere to a unified pre-processing methodology for web pages. Nonetheless, their divergence becomes apparent through the allocation of significance weights to individual keywords and the juxtaposition of identified and unobserved elements in order to provide recommendations of analogous items.

6.1.1 Vector space model (VSM)

It essentially corresponds to a term vector model (TVM) representation. As documented by [72, 73], web pages are encoded as vectors within the vector space model (VSM). Each dimension in these online page vectors signifies the extent of importance attributed to specific phrases within the web page dataset. In simpler terms, the web page for

the “Rs collection” is composed of textual terms. After undergoing a pre-processing procedure, we acquire index phrases or keywords for each web page. Subsequently, weights are assigned to each of these index terms within the collection of web pages. These weights serve to gauge the relevance of an index keyword or the informational content of an internet page. Each web page is preserved in the subsequent format:

$$doc = (kw_1, kw_2, \dots, kw_n)$$

kw_i signifies the weight associated with the term i within the collection of web pages. The parameter n is representative of the total word count within the web-page collection, reflecting the overall informational content. Proficient indexers, equipped with a profound understanding of the content encompassed in web document datasets, have the capacity to make statistical or manual adjustments to the scoring of individual words. Subsequently, following the selection of specific online sites containing textual data that is deemed valuable by users, recommendation systems (RSs) undertake the task of evaluating the similarity between the chosen web pages and those that remain unseen. This evaluation is performed with the purpose of proposing, in ranked order, certain web pages based on their degree of similarity.

6.1.2 Term-weighting schemes in VSM

An “outstanding” indexing keyword, as delineated within the document set by [73], pertains to a term characterized by a notably high discriminative value or weight, thereby minimizing similarities among diverse pages. A rudimentary technique involves employing a keyword or index term weighting system, leveraging the concept of term frequency (tf), which denotes the frequency of occurrence of a phrase within a specific webpage. Nevertheless, this approach bears a limitation since frequently occurring phrases may inadvertently receive elevated weight values across all documents, rendering them suboptimal discriminators. In response to this issue, Sparck Jones has proposed the inverse document frequency (idf) weighting strategy, represented as $\log(TN/Tn)$, where TN denotes the total count of web pages or documents within the collection, and Tn corresponds to the number of documents containing the specific term.

Numerous methodologies for assigning weights in the context of automatic text retrieval were pioneered by Salton and Buckley [74]. According to their framework, the term weighting process comprises three significant components: term frequency, collection frequency, and normalization. Among these configurations, term frequency-inverse document frequency (TF-IDF) stands out as a

prominent example. Current literature, exemplified by research contributions like those of Liu [75], Reed [76], and Greengrass [77], underscores the preeminence of TF-IDF as the most prevalent term-weighting methodology within the vector space model (VSM) domain.

6.1.3 Similarity matching in VSM

The process of assessing the similarity or matching score between a chosen vector and the vectors associated with pages in the collection occurs subsequent to the computation and establishment of term-weight values through a term-weighting scheme (TWS). Following this computation, the websites are presented in descending order of resemblance, with the most similar page to the selected one receiving the highest ranking. This similarity-matching approach is founded on an automated system that assesses the relevance of specific web pages. The precision of the similarity matching function plays a pivotal role in determining the utility of recommendation systems (RSs). Vector space model (VSM) employs five widely recognized similarity-matching functions, which encompass the inner product, cosine similarity, Dice coefficient, Jaccard similarity, and Euclidean distance, when measuring the dissimilarity between page vectors [72, 77].

A significant body of research, as evidenced by [77], has been dedicated to the examination of these functions. Within the realm of vector space model (VSM) analysis, the Cosine similarity function stands out as the prevailing and highly efficient similarity metric. The Cosine similarity function, denoted as $Cosine(Doc, Que)$, serves as a pivotal determinant of the similarity between web pages (Doc) and user queries (Que), defined by the following expression:

$$Cosine(Doc, Que) = \frac{\sum_{i=1}^n KW_{idoc} \cdot KW_{ique}}{\sqrt{\sum_{i=1}^n KW_{idoc}^2 \cdot \sum_{i=1}^n KW_{ique}^2}} \quad (5)$$

Let n represent the count of index terms found within both the unobserved web-page denoted as Doc and the chosen web-page referred to as Que . In the equation provided, KW_{idoc} signifies the weight associated with term i within the content of web-page Doc , while KW_{ique} denotes the weight assigned to the same term i within the content of the selected web-page Que .

6.1.4 Probabilistic models (PM)

Probabilistic models within the realm of information retrieval constitute learning systems that rely on the bag-of-words representation, wherein each document or web page is construed as a vector characterized by term weights. These probabilistic techniques are founded upon robust statistical principles, being fundamentally rooted in the

concept of probabilities. Greengrass [77], for instance, explicated several renowned probabilistic information retrieval models, including the well-regarded Okapi model. The subsequent subsection delves into the intricacies of the Okapi probabilistic model.

6.1.5 Two-poisson probabilistic model (Okapi-best match model)

Robertson et al. introduced a term weighting approach based on the Two-Poisson distribution Models in their seminal work [78]. This technique is notably employed by the Okapi Information Retrieval (IR) system at City London University and has demonstrated remarkable effectiveness, particularly in Text Retrieval Conference (TREC) competitions. To elaborate on the Okapi weighting formula, we introduce the relevant notations. Let TN denote the total number of pages within the collection, and Tn_i represent the count of pages encompassing the phrase i . Furthermore, let $Relq$ denote the total number of online pages relevant to the given page q , and rel_i signify the count of relevant pages containing the phrase i . Moreover, $doctf_i$ stands for the frequency of the term i on page d with a length of $docl$. The sum of term frequencies on a page equates to the page's length. When a page's length is equivalent to $Avgdocl$, it mirrors the average page length within the entire collection. Utilizing these symbols, the Okapi weight for sentence i on webpage d can be formally articulated as follows:

$$w = \frac{doctf_i}{(M_1 \cdot docl)/(Avgdocl) + doctf_i} \cdot w^{(1)}, \text{ and} \quad (6)$$

$$w^{(1)} = \log \frac{(rel_i + 0.5)/(Relq - r_i + 0.5)}{(n_i - rel_i + 0.5)/(N - n_i - Relq + r_i + 0.5)} \quad (7)$$

where M_1 is a constant that typically has a value of 1 or 2 for TREC document collections.

In their investigation, Robertson et al. introduced a novel similarity-matching function as documented in their study [79]. This newly devised function is referred to as “Best Match (BM),” and their experimental inquiry involved the utilization of three distinct formulations of this “Best Match” function, namely BM1, BM11, and BM15. These formulations are instrumental in establishing the “Dot Product” similarity measure [72, 77] between page “d” and the selected page “q” when considering a vector of weights, with BM15 encompassing the “Dot” product alongside a correction factor. Specifically, the BM25 similarity matching function (where “BM25” represents the “Best Match” version 25) is expressed as follows:

$$BM25(Q, D) = \sum_{term\ i \in q} \frac{(CK_1 + 1) \cdot doctf_i}{CK_1 \cdot ((1 - b) + b \cdot \frac{docl}{Avgdocl}) + doctf_i} \cdot \log \frac{(rel_i + 0.5)/(Relq - rel_i + 0.5)}{(n_i - rel_i + 0.5)/(N - n_i - Rq + rel_i + 0.5)} \cdot \frac{qtf_i}{CK_3 + qtf_i} + CK_2 \cdot ql \cdot \frac{(Avgdocl - docl)}{(Avgdocl + docl)} \quad (8)$$

where CK_1 , CK_2 , CK_3 and b are constants that are usually chosen by a trial and error procedure. The simplest form of BM25 by assigning zero values for $Relq$ and rel_i is:

$$BM25(Q, D) = \sum_{term\ i \in q} \frac{(CK_1 + 1) \cdot doctf_i}{CK_1 \cdot ((1 - b) + b \cdot \frac{docl}{Avgdocl}) + doctf_i} \cdot \log \frac{N - n_i + 0.5}{n_i + 0.5} \cdot \frac{qtf_i}{CK_3 + qtf_i} + K_2 \cdot ql \cdot \frac{(Avgdocl - docl)}{(Avgdocl + docl)} \quad (9)$$

TREC tracks, exemplified by TREC-9, have established standard parameter values for retrieval models like the Okapi BM25, wherein typical values for constants like CK_1 and b are 1.2 and 0.7, respectively. However, CK_3 is often assigned values of 0 or 1000, and CK_2 is frequently set to 0. The Okapi BM25 model has stood out as the most effective term-weighting function due to its ability to fine-tune its constants based on characteristics specific to the data collection. Nevertheless, whether or not the relevance judgments in the BM25 equations are applied, the need to adjust these constants necessitates prior knowledge of the relationship between the selected page and the document collection. When these constants are assigned zero values in Okapi BM25, the function essentially reduces to an inner product similarity function within the vector space model (VSM), utilizing an inverse document frequency (IDF) term-weighting scheme.

6.2 Machine learning models

Machine learning algorithms find application in the construction of model-based recommender systems (RSs). The subsequent sections elaborate on distinct machine-learning models employed in this context.

6.2.1 Unsupervised machine learning

This category employs unsupervised machine learning techniques as a key component of data mining research. In the work by [37], these methodologies are extensively elucidated. A predominant focus within this category

pertains to diverse clustering methodologies, which aim to group data entities based on the similarity of their features. Nevertheless, it is worth noting that the heuristic treatment of data attributes may not always represent the most optimal approach for generating clusters that are both robust and devoid of bias, a factor that could have ramifications for the integrity of the information disseminated to end users.

Dimensionality reduction, as discussed in previous works [1, 9, 80], represents a prevalent unsupervised technique employed in the domain of recommender systems. The fundamental objective of dimensionality reduction is to systematically reduce the number of features within a given dataset while striving to retain the maximal amount of pertinent information derived from the original data. In the context of recommender systems, this approach offers notable advantages by ameliorating computational efficiency and diminishing algorithmic complexity. However, it is imperative to acknowledge that the computational cost associated with dimensionality reduction is not uniform across all algorithms; it varies contingent upon the specific methodology employed. Consequently, this variance may pose a substantive challenge when attempting to scale the algorithm to accommodate extensive datasets or to deploy it within real-time applications.

Moreover, dimension reduction methods are designed to preserve a substantial portion of the original information within a dataset while simultaneously reducing the number of features. Nevertheless, this reduction process invariably leads to the loss of some information. The acceptability of this information loss varies depending on the specific context. On one hand, Association Rule Mining (ARM) [81, 82], an unsupervised machine learning technique, uncovers frequent co-occurrence patterns and relationships among items within transactional data, facilitating the identification of item associations. Consequently, items that are commonly acquired together can be suggested as recommendations. On the other hand, an alternative approach involves topic modeling [83, 84], which identifies thematic structures within textual data, such as product descriptions or reviews, and subsequently generates item recommendations based on those identified topics.

Furthermore, collaborative filtering, as elucidated in prior research [85, 86], is a method that endeavors to discern user behavior and preferences to establish affinities among users. This approach subsequently offers recommendations by identifying items that have been favored or purchased by users with analogous tastes. In tandem with user-based collaborative filtering, item-based collaborative filtering techniques, as previously explored [10, 87], present an alternative avenue. The item-based approach proffers suggestions predicated on the resemblance of items to those already endorsed by the user, whereas user-

based collaborative filtering is predicated on the endorsement patterns of comparable users.

6.2.2 Supervised machine learning

The predominant research focus within information retrieval (IR) and content-based recommender systems (CB-RSs) pertains to the practice of ranking retrieved documents based on user feedback. Early investigations in this domain, as illustrated by studies such as [88], employed multimedia (MM) approaches like vector space model (VSM) utilizing techniques such as term frequency-inverse document frequency (TF-IDF) or Okapi-BM25. These methodologies were employed to prioritize retrieved documents, or pages, by assessing their similarity to user-selected pages.

However, a notable limitation was evident in the reliance upon a singular scoring approach, particularly the term-weighting scheme. Such one-dimensional methodology demonstrated inefficiency in the context of creating robust systems. This was chiefly due to scoring techniques like Okapi-BM25, which, while proficient in providing relevant feedback by identifying correct search results, had constraints in delivering comprehensive feedback [89–91].

According to [92], there are three types of LTR approaches: (1) the pointwise technique, (2) the pairwise method, and (3) the listwise method. Assessment procedures of the loss function or fitness function determine these groups. The learning instance in the pointwise technique is every single object (rated web-page-unseen web-page pair). Linear regression (LR) [93], boosting [94], gradient boosted regression trees (GBRT or MART) [95, 96], and random forest (RF) [97] are examples of pointwise techniques. The pairwise technique considers the learning instance to be a pair of objects (two rated web-page-unseen web-page pairs for the same user). RankNET (rank neural Net) [98], RankBoost and SVMRank (rank support vector machine) [99] are examples of pairwise techniques. The learning instances in the listwise technique are the complete obtained list of items (the list of rated web-page-unseen web-page pairings for each user). ListNET (listwise neural net) [100] and RankGPES [101] are two examples of listwise techniques.

Whereas listwise techniques have been demonstrated to outperform pointwise and pairwise methodologies in terms of accuracy [100], the desire to increase the performance of LTR approaches has motivated researchers to create hybrid methods. Sculley, for example, introduced a method (CoRR) that combined linear regression (point-wise) with support vector machine (pair-wise) [102]. LambdaRank and LambdaMART are two more hybrid systems that integrate pair-wise and list-wise algorithms. [103] was also presented. LambdaRank is built on RankNET, whereas

LambdaMART is a boosted tree derived from LambdaRank. Both LambdaMART and LambdaRank outperformed Mohan et al.'s technique on the Yahoo! LTR Challenge [104] in terms of IR accuracy. Thus, combining listwise and pointwise approaches has shown to be fruitful. Muahmmed and Carman did tests combining listwise and pointwise random forest (Hybrid RF), demonstrating that their hybrid outperformed other pointwise and listwise RF in computational run-time and accuracy [105].

7 Towards robust recommendation systems

In this section, we delve into a range of factors contributing to biases in recommender system (RS) evaluations. These factors encompass issues such as the dissemination of inaccurate product information and the presence of fraudulent products, both of which can influence consumer perceptions. We commence by examining the various problems stemming from bias in RSs, along with potential mitigation strategies involving trust-aware recommender systems (RSs). It is important to note, however, that when the integrity of the system's information security is compromised due to malicious software and spyware, trust-aware RSs may encounter challenges in ensuring equitable recommendations. Furthermore, the RS database can be vulnerable to security threats, including SQL injections. Additionally, some users may lack a clear understanding of the credibility of their network connections, as exemplified by teenagers on social media platforms who tend to form connections with unfamiliar individuals. This dynamic within the "Friends" zone of social media can be viewed as the trust-aware zone for RSs. Consequently, the influence of these connections within trust-aware collaborative filtering RSs on social media platforms may inadvertently introduce biases into the decision-making processes of teenagers.

7.1 Bias-based attacks on recommender systems

Shilling attacks, as highlighted in the study by [106], represent a significant source of bias within recommender systems (RSs). In these attacks, both individuals and automated bots emulate legitimate user behaviors by employing deceptive user accounts. These actors then manipulate RS feedback mechanisms through actions such as boosting (push) or diminishing (nuke) feedback for specific pieces of content. Chirita et al. [107], in their research, identified three metrics for the detection of shilling attacks utilizing user ratings.

The first metric, known as the "number of prediction difference," involves a comparison of the system's predictions before and after the removal of a specific user.

Substantial disparities in prediction outcomes before and after user removal indicate a bias in the RS towards that particular user.

The second metric, termed "standard deviation in user's ratings," focuses on the variability in ratings provided by a user. This measure assists in identifying suspicious patterns in user behavior that may be indicative of shilling attacks.

This metric assesses the disparity between user ratings for items and the average ratings provided by the system. To compute these discrepancies, the standard deviations of user ratings are employed. These standard deviation values gauge the impact of individual user biases. Another significant statistic for bias detection in recommender systems (RSs) is the "degree of agreement with other users," which represents the mean of these standard deviations.

Instances of fraudulent account usage, as well as the presence of "gray sheep users (GSUs)," as introduced by [108], or groups of users with unconventional views on RSs products, can introduce bias into the system. Distinguishing between genuine feedback and possible manipulative tactics, such as shilling attacks conducted through fake accounts, poses a challenge. Consequently, RSs developers have introduced the concept of trust-aware RSs as an innovative approach to constructing reliable RSs. In this model, users are granted the ability to assign trust values to the recommendations provided by their peers, enhancing the discernment of reliable feedback. Noteworthy examples of this type of RS include platforms like Epinions and certain social networks. A more detailed discussion of trust-aware RSs will follow in the subsequent section.

7.2 Trust-aware recommender systems

The leading methodology for the development of recommendation engine applications, aimed at mitigating bias arising from multiple adversarial activities involving counterfeit user accounts, is the trust-aware approach. Within trust-aware recommender systems (RSs), the process involves the provision of item suggestions for the target active user, a task carried out either by trusted individual users or collectively by all trusted users participating in the RSs. The existing scholarly literature highlights the utilization of three established trust-aware approaches as documented in previous research [49, 109]: 1) tidal trust, 2) trust-based collaborative filtering, and 3) the "trust-based weighted mean" technique involves summarizing various strategies as outlined below. It presupposes the existence of a trust weight denoted as $W_{a,u}$ between an active user, denoted as a , and another user, denoted as u . This trust weight, $W_{a,u}$, quantifies the level of trust that the active user

a “places in the ratings provided by user *u*.” Moreover, the method assumes that user *u* assigns a rating, represented as $r_{u,i}$, to an item of information denoted as *i*. Consequently, employing the Trust-based Weighted Mean technique, the anticipated rating, denoted as $PR_{a,i}$, for item *i* by the active user *a* can be determined as elaborated in [109].

$$PR_{a,i} = \frac{\sum_{u \in R^T} W_{a,u} \cdot r_{u,i}}{\sum_{u \in R^T} W_{a,u}} \quad (10)$$

Let R^T denote the set of users who have provided ratings for a specific item, denoted as *i*, while also satisfying a predefined threshold criterion based on their trust values ($W_{a,u}$). In the context of collaborative filtering techniques that incorporate trust-based mechanisms, the approach known as trust-based collaborative filtering represents a notable example, ranking as the second trust-aware technique in this category. As elucidated in the work by [110], the following equation serves as a pivotal tool for estimating the anticipated rating assigned by an active user, denoted as *a*, for a particular information item, marked as *i*:

$$PR_{a,i} = \bar{r}_a + \frac{\sum_{u \in R^T} W_{a,u} (r_{u,i} - \bar{r}_u)}{\sum_{u \in R^T} W_{a,u}} \quad (11)$$

In the context of this study, let $\bar{r}_a$ represent the mean ratings given by user *a* to other items, and $\bar{r}_u$ denote the mean ratings provided by user *u* for item *i*. An alternative approach known as “TidalTrust” is introduced as the third trust-aware method. TidalTrust bears similarity to the trust-based weighted mean approach, with a notable distinction lying in its consideration of the cohort of individuals trusted by active users *a*. Consequently, within the framework of Eq. 10, the determination of trust-weight values between an active user *a* and users *u* takes on the following form:

$$W_{a,u} = \frac{\sum_{v \in WOT^+(a)} W_{a,v} \cdot W_{v,u}}{\sum_{v \in WOT^+(a)} W_{a,v}} \quad (12)$$

Let $WOT^+(a)$ represent the ensemble of individuals with whom user *a* has established trust relationships that exceed a specified threshold. Moreover, consider user *u* as a “friend-of-friend” of user *a* via the intermediary user *v*. The trust weights assigned to the interactions between user *a* and *v* and between user *v* and *u* are denoted as $W_{a,v}$ and $W_{v,u}$, respectively.

On the contrary, a notable limitation observed in the prevalent methods employed for product rating within recommender systems (RSs) warrants attention. This limitation predominantly restricts the effectiveness of prior Trust-Aware approaches to managing explicitly endorsed user ratings. Notably, established scholars such as [111, 112] have advanced the notion that through the adept utilization of collaborative filtering (CF) and Similarity-

based techniques, it is feasible to address the challenge posed by implicit ratings, both at the item-level and profile-level. By doing so, the numeric values associated with CF-based ratings, as well as the correlation or similarity metrics computed between information items or user profiles, emerge as viable alternatives to the conventional user-provided rating values employed in earlier methodologies.

Furthermore, Zeigler and Golbeck [113] conducted an investigation into the intricate relationship between the similarity of user ratings and the degree of trust within recommender systems (RSs). Their research revealed a noteworthy association between trust levels and the assessment of items within RSs. Consequently, it is tenable that the integration of trust-aware methodologies offers a robust defense mechanism against biases induced by fraudulent or deceptive activities, such as shilling and gray sheep attacks. Consequently, prominent social media platforms and established online review websites have demonstrated a preference for incorporating diverse trust-aware strategies into their recommendation systems. Nevertheless, authentic and reliable users within RSs may exhibit variations in their trust levels, displaying trust in certain domains while exhibiting skepticism or partiality in others. This variation has led some social networking platforms to adopt the circle-based trust-aware approach in their recommendation engines. This method entails the categorization of trust levels into distinct groups; for instance, a user may place trust in another user’s recommendations pertaining to movies but maintain skepticism when it comes to fashion-related items, thus exemplifying the nuanced nature of user trust.

8 A comparison between some well-known recommender system methods

The following table shows a comparison among the most recent methods proposed for recommender systems. This table provided the methods with their source code for research reproducibility with various recommendation systems datasets. Microsoft’s research team has developed these methods from the literature (Table 1).

9 Future research implications

Technological advancements, changes in scenarios, and changes in user expectations offer significant opportunities for RSs; however, existing RS data, algorithms, and evaluation techniques are also challenged. Researchers must evaluate the limitations and potential extensions of current RS research from various perspectives to respond to these emerging trends effectively. In the following subsections,

Table 1 A list of some recommender system methods which are available for research reproducibility

Algorithm	Type	Description	Strengths	Weaknesses	Code example
Alternating least squares (ALS) [114]	Collaborative filtering	Matrix factorization technique optimised for scaling and dispersed computing capacity in huge datasets with explicit or implicit feedback. It is functional in the PySpark as framework	(1) Scalability: ALS can be efficiently parallelized, making it suitable for large-scale recommendation systems. (2) Sparsity Handling: ALS handles sparse matrices well, which is common in user-item interactions	(1) Cold Start Problem: ALS struggles with new users or items that have very few interactions. (2) Overfitting: If regularization is not properly tuned, ALS can overfit the data, especially when there is noise or bias in the ratings.	Quick start
BPR [115]	Collaborative Filtering	With implicit consumer feedback, a matrix factorization technique forecasts item rating. It operates only in the CPU system	(1) Effective for implicit feedback. (2) Scalability: It can handle large datasets and scales reasonably well with proper optimization. (3) Ranking-Oriented: Its focus on ranking items rather than predicting ratings.	(1) Cold Start Problem (new users/items with little or no interaction data). (2) The performance depends heavily on how the negative samples (uninteracted items).	Quick start
BiVAE [116]	Collaborative Filtering	Models for generating dyadic data (for example, user-item engagements). It is compatible with both the CPU and GPU platforms	(1) BVAE can capture personalized interactions between users and items more effectively by directly models the user and item latent spaces. (2) by using a bilateral approach, BVAE can incorporate richer and more diverse data for both users and items. (3) The generative nature of BVAE allows for generating potential future interactions. (4) BVAE's variational inference framework provides uncertainty estimates for the latent variables. (5) BVAE can be extended to handle multimodal inputs with additional information like textual descriptions or images associated with items.	(1) Limited to Sequential Data. (2) Limited Modeling of Long-Term dependencies. (3) The features learned are not easily interpretable.	Quick start
Caser [117]	Collaborative filtering	Convolution-based technique attempting to exploit both the user's broad interests and consecutive patterns. It is compatible with both the CPU and GPU platforms	(1) Sequential pattern extraction: Caser can effectively capture sequential patterns and dependencies in user behavior. (2) Local and Global Dependencies: Caser's combination of horizontal and vertical filters enables it to capture both short-term (local) and long-term (global) user preferences. (3) Scalability: Convolutional operations are highly parallelizable, making Caser scalable to large datasets. (4) Personalized Recommendations: Caser excels at personalizing recommendations by learning user-specific patterns from historical data. (5) Adaptability to Different Length Sequences: Caser can handle sequences of varying lengths which allows it to flexibly adjust to both short and long interaction histories. (6) Ease of Training.	(1) Limited to Sequential Data. (2) Limited Modeling of Long-Term dependencies. (3) Interpretability: the features learned by Caser functions are not easily interpretable. (4) Sequential Bias: Caser emphasizes the sequential nature of interactions, which can sometimes introduce bias.	Quick start

Table 1 (continued)

Algorithm	Type	Description	Strengths	Weaknesses	Code example
DKN [118]	Content-based filtering	For giving headline or content suggestions, a deep learning technique is used with a knowledge graph and article embeddings. It is compatible with both the CPU and GPU platforms	Knowledge-Aware Recommendations: By leveraging a knowledge graph, DKN can incorporate external information that improves the recommendation quality. (1) Combining content and knowledge: DKN effectively integrates both the textual content of items and external knowledge from a graph. (2) Enhanced user understanding: The knowledge graph provides more context to user behavior. (3) Attention mechanism: The attention mechanism improves the model's ability to focus on the most relevant aspects of the content. (4) Personalization: DKN learns user preferences from historical interactions.	(1) Complexity: DKN is more complex than traditional recommendation models due to the need for both textual and knowledge graph embeddings, multi-channel CNNs, and attention mechanisms. (2) Knowledge graph dependency: The performance of DKN depends heavily on the quality and coverage of the knowledge graph. (3) Data preparation: DKN requires significant preprocessing, including extracting entities from text and linking them to the knowledge graph. This preprocessing step can be time-consuming and requires access to a well-maintained knowledge graph. (4) Cold start problem.	Quick start
xDeepFM [119]	Hybrid	An algorithm developed using deep learning for implicit and explicit feedback based on user and item attributes. It performs across the CPU and GPU platforms	(1) Combining Explicit and Implicit Feature Interactions. (2) The CIN component allows the model to learn feature interactions at different levels, providing a more structured and hierarchical approach to understanding the data. (3) The DNN component in xDeepFM allows the model to generalize better to unseen data. This reduces overfitting and improves the model's performance on large and diverse datasets. (4) The model is highly parallelizable and can be efficiently trained on large datasets. (5) xDeepFM is highly flexible and can be easily adapted to different types of tasks.	(1) xDeepFM is more complex than traditional models. (2) The performance of xDeepFM is sensitive to hyperparameter settings, such as the number of layers and units in the DNN. (3) DeepFM requires a substantial amount of data to train effectively.	Quick start
FastAI Embedding Dot Bias [120]	Collaborative Filtering	An approach that incorporates embeddings and biases for users and things. It is compatible with both CPU and GPU platforms	The use of embeddings and dot products makes it fast to train, even on large datasets with many users and items. (1) The addition of bias terms improves prediction accuracy over pure matrix factorization models without biases. (2) EmbeddingDotBias scales well to large datasets, making it suitable for real-world recommendation systems.	(1) The model only captures linear interactions between users and items (via the dot product). (2) Cold Start Problem. (3) EmbeddingDotBias focuses on pairwise interactions (i.e., user-item interactions). This limits the model's ability to leverage richer side information.	Quick start

Table 1 (continued)

Algorithm	Type	Description	Strengths	Weaknesses	Code example
LightFM or Hybrid Matrix Factorization [121]	Hybrid	A hybrid matrix factorization method that considers implicit as well as explicit feedback. It is functional in the CPU system	(1) LightFM's ability to incorporate both collaborative filtering and content-based filtering allows it to overcome the limitations of either approach when used alone. (2) LightFM provides a range of loss functions (e.g., logistic loss, BPR, WARP) that can be chosen based on the task at hand. (3) The model is well-suited for dealing with both implicit and explicit feedback. (4) By incorporating user and item features, LightFM mitigates the cold start problem. (5) LightFM is optimized for large-scale datasets and can efficiently handle recommendation tasks with millions of users and items.	(1) LightFM can capture some complex patterns through embeddings, it may miss non-linear interactions. (2) The model cannot automatically discover complex feature relationships like deep learning models. (3) Cold Start Limitations. LightFM does not model higher-order interactions explicitly. For instance, interactions between multiple users, items, or features. (4) Choosing the right loss function, embedding dimensions, and regularization parameters can require extensive hyperparameter tuning.	Quick start
LightGBM [122]	Content-Based Filtering	In content-based problems, the Gradient Boosting Tree technique is used for rapid learning and minimal memory utilization. It operates on the CPU and GPU using the PySpark framework	(1) LightGBM is designed to handle very large datasets and high-dimensional data efficiently. (2) The histogram-based approach to bin continuous features into discrete values significantly reduces memory consumption, making it suitable for big data scenarios. (3) It can handle categorical features natively, without the need for encoding, which is both time-saving and memory-efficient. (4) It supports distributed training, which allows it to scale across multiple machines, enabling efficient training on large datasets. (5) LightGBM supports GPU acceleration, which can significantly reduce training times for large datasets. (6) LightGBM can be used for a wide variety of machine learning tasks, including classification, regression, ranking, and even recommendation systems.	(1) LightGBM is more prone to overfitting, especially when the model is not properly regularized or when the dataset is small. (2) LightGBM's performance is highly dependent on hyperparameter tuning. (3) On smaller datasets, LightGBM's leaf-wise growth strategy may lead to overfitting more easily compared to level-wise methods like XGBoost. (4) LightGBM models are generally less interpretable than simpler models like linear regression or decision trees. (5) LightGBM can sometimes be tricky to install and set up, especially on certain platforms.	Quick start
GeoIMC [123]	Hybrid	Matrix completion technique that uses Riemannian conjugated gradients optimisation and a geometrical approach to account for user and item properties. It only works in the CPU system	Integrating geographical information can significantly enhance the relevance of recommendations by considering location-based preferences and behaviors. (1) GeoIMC provides context-aware recommendations by incorporating location-based features. (2) GeoIMC can improve the accuracy of matrix completion and recommendation models. (3) The approach provides valuable insights into regional or location-specific trends and preferences.	(1) Integrating geographical features into matrix completion can add complexity to the model and the training process. (2) Using geographical information raises privacy concerns, as users may be uncomfortable with their location data being used for recommendations. (3) Depending on the implementation, incorporating geographical features may impact the scalability of the model.	Quick start

Table 1 (continued)

Algorithm	Type	Description	Strengths	Weaknesses	Code example
GRU4Rec [124]	Collaborative Filtering	Using recurrent neural networks, this system recognizes both long-term and short-term customer preferences. It operates in a CPU/GPU system	(1) GRU4Rec excels at capturing sequential patterns and temporal dependencies in user interactions. (2) The GRU architecture is designed to handle long-term dependencies effectively, which helps in learning from both recent and past interactions in the user's sequence. (3) GRU4Rec suitable for large-scale recommendation tasks. (4) The model can be adapted to various recommendation scenarios, including predicting the next item in a sequence, ranking items, or personalized recommendations based on user history.	(1) GRU4Rec may not fully leverage additional contextual information (e.g., item metadata or user demographics). (2) GRU4Rec may struggle with data sparsity issues, especially when user interaction sequences are short or when there are few interactions for certain items. (3) Handling very large datasets or sequences with GRU4Rec may require significant computational resources and optimization.	Quick start
Multinomial VAE [125]	Collaborative Filtering	Predicting user/item responses using a generative model. It operates in a CPU/GPU system	Handling Categorical Data: Multinomial VAE is specifically designed to handle data with categorical distributions, making it suitable for tasks involving counts, categorical features, or discrete data. (1) Multinomial VAE can be used to generate new samples that are similar to the training data. (2) VAE learns a latent space representation that captures the underlying structure of the categorical data. (3) The use of a multinomial distribution allows for flexible modeling of categorical data.	(1) It need to handle discrete data and multinomial distributions. (2) Modeling categorical data with a multinomial distribution can be challenging if the data has a very large number of categories or if the categories are imbalanced. (3) Designing the appropriate loss function for multinomial VAE can be more challenging compared to continuous data.	Quick start
LSTUR [126]	Content-Based Filtering	Long- and short-term user interest modelling is used in a neural recommendation system for proposing news reports. It operates in a CPU/GPU system	(1) The model effectively captures both recent and historical user behavior. (2) The model provides more personalized and contextually relevant recommendations. (3) The architecture is flexible and can be adapted to various types of user interaction data and recommendation tasks.	(1) The model architecture can be complex, requiring careful tuning of hyperparameters and design choices. (2) The model requires sufficient historical interaction data. Sparse data may lead to suboptimal performance.	Quick start
NAML [127]	Content-Based Filtering	With attentive multi-view learning, a neural recommendation system is used to propose news topics. It operates in a CPU/GPU system	(1) The attention mechanism helps to focus on the most relevant features and views, improving the model's ability to capture important patterns and relationships. (2) Multi-view learning allows for more nuanced and personalized recommendations.	(1) Increased training and inference times due to the involvement of multiple neural network components and attention mechanisms. (2) Integrating and processing multiple views requires careful handling and alignment of data, which can be challenging and require additional preprocessing. (3) Efficient implementation and optimization are necessary to manage computational resources. (4) The approach requires access to diverse types of data views, which may not always be available.	Quick start

Table 1 (continued)

Algorithm	Type	Description	Strengths	Weaknesses	Code example
NCF [128]	Collaborative Filtering	Deep learning technique with improved performance for implicit user/item interaction. It operates in a CPU/GPU system	(1) NCF offers superior performance at capturing complex, non-linear relationships between users and items. (2) NCF can be adapted to various types of recommendation tasks, including implicit feedback (e.g., clicks, views). (3) NCF often outperforms traditional matrix factorization methods in terms of recommendation accuracy. (4) A hybrid approach that balances the simplicity of linear interactions with the power of nonlinear modeling.	(1) NCF models are more computationally expensive to train and require more hyperparameter tuning. (2) For large-scale datasets, training and inference times can become prohibitively expensive. (3) Neural networks do not provide clear interpretations of how specific recommendations are made.	Quick start
NPA [129]	Content-Based Filtering	Personalised attention neural network recommendation method for proposing news documents. It operates in a CPU/GPU system	(1) The model can account for unique preferences and behaviors, leading to more accurate and personalized recommendations. (2) Attention mechanisms provide some level of interpretability in recommendation systems. (3) Personalized attention allows the model to focus on the most relevant parts of complex, multi-faceted data, such as sequential interactions, contextual information, and multi-modal features (e.g., text, images). (3) The architecture is flexible and can be adapted to various types of recommendation tasks.	(1) The introduction of personalized attention mechanisms adds complexity to the model, increasing both the computational cost of training and inference. (2) Since the model generates attention weights by using past user interactions, it may struggle with sparse data, especially for new or infrequent users. (3) Optimization techniques, such as sampling or parallelization, may be needed to handle large-scale data efficiently. (4) Cold Start Problem	Quick start
NRMS [130]	Content-Based Filtering	A neural recommendation algorithm with multi-head self-attention is applied to propose news items. It functions in a CPU/GPU context	(1) Self-attention allows the model to capture dependencies between interactions that are far apart in the sequence. (2) The multi-head mechanism enables the model to capture multiple types of relationships between user interactions and items simultaneously. (3) The architecture is highly versatile and can be applied to various types of recommendation tasks.	(1) The transformer-based architecture with multi-head self-attention is computationally expensive. (2) The model requires a significant amount of data to learn meaningful representations effectively. (3) Cold Start Problem. (4) The complexity of multi-head self-attention can make it difficult to understand exactly why certain recommendations were made.	Quick start

Table 1 (continued)

Algorithm	Type	Description	Strengths	Weaknesses	Code example
Next Item Recommendation (NextItNet) [131]	Collaborative Filtering	Algorithm for capturing pattern sequences that use dilated convolutions and residual nets. It takes into account both user/item interactions and features. It operates in a CPU/GPU system	(1) Temporal Awareness: Next item recommendation captures the sequential nature of user interactions, allowing for more dynamic and timely recommendations that reflect recent behavior patterns. (2) Personalization: These models are highly personalized because they base recommendations on the unique interaction history of each user, rather than relying solely on aggregated user preferences. (3) Context Sensitivity: When combined with contextual information, next item recommendation can adapt to the user's situation (e.g., time or location), making the predictions more relevant and context-aware. (4) Flexibility: The approach can be applied to a variety of domains, from recommending products and movies to predicting the next song in a playlist or the next article to read.	(1) Cold Start Problem: Like many personalized models, next item recommendation struggles with the cold start problem, where there is insufficient data for new users or items. Solutions like hybrid models or content-based filtering can mitigate this issue. (2) Computational Complexity: Sequential models, especially transformer-based architectures, can be computationally expensive to train and deploy, particularly for long interaction sequences or large datasets. (3) Data Sparsity: In domains with sparse user-item interactions, it can be challenging to model sequences effectively, as there may be insufficient data to learn meaningful patterns. (4) Overfitting to Short-Term Patterns: While the models are effective at capturing short-term user behaviors, they can sometimes overfit to recent interactions and fail to generalize to longer-term preferences. Balancing short-term and long-term patterns remains a challenge.	Quick start
Restricted Boltzmann Machines (RBM) [132]	Collaborative Filtering	Algorithm that uses neural networks for learning the base probability distribution for implicit or explicit user/item interaction. It operates in a CPU/GPU system	(1) Efficient Inference: RBMs have a relatively simple structure, allowing for efficient inference and sampling of hidden and visible units. (2) Unsupervised Learning: RBMs can learn useful features from unlabeled data, making them valuable in scenarios where labeled data is scarce. (3) Good for Collaborative Filtering: RBMs have shown strong performance in collaborative filtering, particularly when user-item interactions are sparse, as they can capture underlying latent factors in the data.	(1) Difficult Training: Training RBMs can be challenging due to the need for approximations in the learning process (e.g., using Contrastive Divergence). This can lead to slow convergence or suboptimal solutions. (2) Scaling Issues: RBMs do not scale well to large datasets or deep architectures due to the computational cost of sampling and learning. (3) Limited Expressiveness: Compared to more recent deep learning models, RBMs have limited expressiveness and are often outperformed by newer architectures like variational autoencoders (VAEs) or generative adversarial networks (GANs). (4) Sensitivity to Hyperparameters: RBMs are sensitive to hyperparameters, such as learning rate, number of hidden units, and the number of Contrastive Divergence steps	Quick start

Table 1 (continued)

Algorithm	Type	Description	Strengths	Weaknesses	Code example
Riemannian Low-rank Matrix Completion (RLRMC) [133]	Collaborative Filtering	To forecast user/item relationships, a matrix factorization approach with Riemannian conjugate gradients optimisation and low memory usage is used. It works in the CPU system	(1) Efficiency: RLMC methods are more computationally efficient compared to traditional low-rank matrix completion approaches, as they explicitly incorporate the low-rank constraint into the optimization process. This leads to faster convergence and reduced computational overhead. (2) Scalability: Due to the dimensionality reduction achieved through low-rank approximations, RLMC methods can scale to larger datasets, making them suitable for applications like recommender systems, where data sparsity is an issue but the matrix is often large. (3) Accurate Recovery: The Riemannian framework allows for more accurate recovery of the missing matrix entries, especially when the underlying data truly follows a low-rank structure. The optimization respects the intrinsic geometry of the low-rank manifold, leading to better results than methods that operate in the full Euclidean space. (4) Geometric Intuition: By leveraging the geometric properties of the data, RLMC models can capture more complex relationships within the data compared to traditional linear methods. This is particularly beneficial in applications where data lies on non-linear manifolds.	(1) Complexity of Implementation: RLMC methods are more complex to implement compared to traditional matrix completion techniques. They require specialized knowledge of Riemannian geometry and optimization on manifolds, making them less accessible for practitioners unfamiliar with these concepts. (2) Limited Applicability: While powerful for problems with an inherent low-rank structure, RLMC may not be suitable for datasets that do not adhere to this assumption. If the true data does not follow a low-rank model, the performance may degrade. (3) Algorithmic Overhead: Although RLMC methods are more efficient in terms of memory and computations due to the low-rank assumptions, the Riemannian optimization algorithms themselves can have higher overhead per iteration compared to simpler algorithms, such as standard gradient descent. (4) Sensitivity to Rank Estimation: The rank of the matrix is typically an input parameter for RLMC methods. If the rank is misestimated, it can lead to suboptimal recovery or unnecessary computational expense. Proper rank estimation is crucial for good performance.	Quick start
Simple Algorithm for Recommendation (SAR) [134]	Collaborative Filtering	A technique based on similarity for implicit user/item assessment. It is functional in the CPU system	(1) Simplicity: Easy to implement. (2) Interpretability: Recommendations are based on the preferences of similar users, making the logic easy to understand. (3) No Need for Item Metadata: Purely based on user behavior, so no additional item information is needed.	(1) Cold Start Problem: Struggles with new users or items with little interaction data (since similarity cannot be computed). (2) Sparsity: If the user-item matrix is too sparse, it may be difficult to find similar users.	Quick start

Table 1 (continued)

Algorithm	Type	Description	Strengths	Weaknesses	Code example
Self-Attentive Sequential Recommendation (SASRec) [135]	Collaborative Filtering	A sequential system of recommendations that utilises transformers. It is compatible with both CPU and GPU systems	(1) Long-Range Dependencies: Self-attention allows the model to capture long-range dependencies between items in the sequence. This is particularly useful when earlier items in the sequence (e.g., a purchase several interactions ago) have a strong influence on the next item. (2) Parallelization: Unlike recurrent neural networks (RNNs), which process sequences step by step, self-attention can process the entire sequence in parallel, making it more computationally efficient and faster to train. (3) Flexibility: The model is flexible and can capture complex, nonlinear patterns in the user's interaction history. It does not assume a fixed order of dependencies like traditional sequential models (e.g., RNNs or Markov models). (4) Adaptive attention: The attention mechanism is dynamic, allowing the model to focus on the most relevant past interactions at each time step.	(1) Position Sensitivity: Although positional encodings are used, self-attention mechanisms are fundamentally position-agnostic, which means they may struggle with tasks that require strict adherence to sequential order. (2) Data Hunger: Self-attentive models require a large amount of data to train effectively. They need extensive interaction histories to learn meaningful patterns and may not perform well with sparse data. (3) High Computational Cost: The self-attention mechanism has a time complexity. This makes it expensive for long sequences, though this can be mitigated with techniques like sparse attention or other optimizations. (4) Cold Start Problem: Similar to other collaborative filtering methods.	Quick start

¹ BPR: CornacBayesian Personalized Ranking. BiVAE: CornacBilateral Variational Autoencoder. Caser: Convolutional Sequence Embedding Recommendation. DKN:Deep Knowledge-Aware Network. xDeepFM: Extreme Deep Factorization Machine. FAST: FastAI Embedding Dot Bias. LSTUR: Neural Recommendation with Long- and Short-term User Representations. ² NAML: Neural Recommendation with Attentive Multi-View Learning. NCF: Neural collaborative filtering. NPA: Neural Recommendation with Personalized Attention. NRMS: Neural Recommendation with Multi-Head Self-Attention

we will outline new opportunities for future research in the area of recommendation systems.

9.1 Developing enhanced trust-aware techniques

Cold start problems are the challenges that arise when recommending new users or things with no or little interaction history [7, 8]. These problems pose significant obstacles to the creation of tailored recommendations. Using an uncertainty measure $U(v)$, one can choose a component from a group of components $[V]$ to maximize the predicted reduction in uncertainty within the model.

$$v^* = \arg \max_{v \in V} \mathbb{E}_v[U(v)] \quad (13)$$

A Trust-Aware Recommender System (RS) uses relationships of trust among users to enhance the accuracy of recommendations, particularly in the case of cold-start problems and content discovery. Multimodal data are currently being used to enhance historical interactions. Several studies have used deep learning to integrate

modalities like text, voice, and visuals to improve RSs' comprehension and prediction of user interest [65, 66]. Several researchers utilize social network data to enhance the representation of user features [17, 31, 40]. Furthermore, studies that use environmental and sensor data to create contextual signals that enhance RS's utility and application [63]. To provide reliable real-time recommendations, Hybrid trust-aware systems sometimes require the integration of various data sources. In hybrid recommender systems, deep learning has greatly enhanced the integration of a number of pieces of information, such as content and context. The study by Gat et al. [136] uses graph embeddings and contextual data to improve suggestions, while the study by Polignano et al. [137] integrates user-item interaction with contextual data to enhance suggestions. The majority of the literature have considered the following techniques in the context of developing enhanced trust-aware RSs [7, 8, 40, 136]:

1. **Trust propagation:** Trust propagation is the process of distributing trust scores among users in a network. The reliability of users with whom one has indirect

connections can be determined if one has trust relationships with other users. Based on the transitive property of trust, user *A* may also trust user *C*, although to a limited degree, if user *A* trusts user *B* and user *B* trusts user *C*. Despite the fact that user *A*'s preferences may not be directly known, recommendations can still be made based upon this implicit trust.

2. **Trust-based filtering:** Through trust-based filtering, recommendations from members of a trust network take precedence over recommendations from a broader user base. In contrast to typical collaborative filtering, which relies on the preferences of comparable users regardless of their relationship, trust-based filtering relies on the opinions of users who are explicitly trusted by the active user.
3. **Hybrid trust-aware models:** In hybrid models, multiple recommendations techniques are combined to maximize their strengths. Trust data, for example, can be used to complement collaborative filtering by adding a layer of reliability, ensuring that recommendations are not merely based on statistical similarities but also on relationships of trust. By incorporating trust into content-based methods, users can receive recommendations that are not only similar to previous favorites, but also endorsed by people they trust.
4. **Social trust networks:** In social networks, trust relationships are naturally captured through connections, likes, follows, and interactions. By analyzing this data, recommender systems are able to determine which users trust each other. Using social trust to enhance recommendations is particularly important in cold-start scenarios, where users may not have much interaction data but do have a number of social connections.

Although an effort has been made to resolve the cold start and data sparsity problems, there are still a number of important obstacles to overcome. First of all, traditional data modalities, such as text and images, tend to remain static over time due to their characteristics or content. It is unable to respond to the interactive and real-time requirements of users due to its static nature. A second change in how people interact and obtain information is due to new scenarios and technologies such as mixed reality and brain-computer interfaces [138]. There is still a need for additional research in order to fully understand and record user actions and preferences in virtual worlds using physiological signals, such as eye movements and EEGs.

9.2 Integration of collaborative filtering and machine learning

The combination of CF with machine learning (ML) content-based filtering has generated hybrid systems that are more resilient [13, 21, 35]. The use of convolutional neural networks (CNNs), for example, has been demonstrated to enhance the accuracy of recommendation when features from item photos are extracted and merged with CF information [68, 139]. It is particularly relevant in the field of e-commerce. Machine learning (ML) enables these systems to combine user-item interaction data (CF) with other information, such as user profiles, item descriptions, and contextual information (such as time or location). Through the incorporation of machine learning, CF models are being improved in terms of their explainability and fairness. In addition to providing insight into the recommendations made for certain goods, attention mechanisms increase recommendation accuracy and boost user confidence [69, 70]. A ML-driven technique is also being investigated to ensure that recommendations are equitable across various demographics of users in order to prevent biases that may arise from the data. However, there are still many issues that need to be addressed, such as making sure that the model is transparent, fair, and able to be generalized across a wide range of applications [14, 15, 140]. As these models become more complex, sophisticated training and inference techniques are required to maintain real-time recommendation capabilities. Integration and management of contextual data (such as location, device, and current activity) into the recommendation process can be a challenging task. It is important for systems to remain relevant over time since the context of users can change over time. The management of large user populations and large data sets in real time is a significant challenge. There is a challenge in balancing the need for personalization with user privacy and data protection laws.

9.3 Addressing proprietary knowledge gaps

In order to maintain a competitive advantage, companies can develop proprietary algorithms, data handling methods, or system architectures [14, 58]. As a result of personalized suggestions from recommender systems (RS), client satisfaction increases and revenue growth is stimulated. It is integral to the business strategies of large companies, such as Amazon, Netflix, and Spotify, which are based on RS. Amazon claims that their RS generates 35% of its income [61], while Netflix claims that its RS accounts for a significant portion of its 33.7 billion in sales and its effectiveness in retaining customers. Each application domain, such as e-commerce or e-learning, presents its own set of

constraints for recommender systems (RS). In contrast to e-learning systems which must take into account a variety of learning styles and educational goals, e-commerce systems must adapt to changing inventories and consumer trends [41, 49, 103]. In light of these domain-specific challenges, RS must be flexible, scalable, and technically sound, as well as ethically sound and morally sound for enterprises of all sizes. Technical limitations, challenges with adaptability, and ethical considerations are only some of the limitations associated with RS [34, 59, 61]. Real-world implementation of RS is more challenging, particularly for small- and medium-sized businesses. As a result of a lack of funding and technical expertise, it can be challenging to implement advanced RS, which can lead to problems with scalability and adaptability to rapidly changing market conditions [34, 41, 49, 103]. Additionally, issues with review data, including its veracity, quality, and manipulation potential, complicate the application of RS. The substantial amount of data collection necessary for individualized suggestions raises significant privacy concerns, especially regarding consent and data security. Data management for user reviews also poses privacy concerns since businesses must strike a balance between user privacy protection and personalization of suggestions.

9.4 Mitigating bias and enhancing serendipity

Popularity Bias is one of the most common forms of bias in RSs, characterized by a tendency to prefer popular items over less popular ones. Based on the user's past behavior, User Bias may exclude novel or diverse options. Through content features, such as demographic information or product attributes, Content Bias can be introduced. In terms of improving serendipity and mitigating bias, the existing body of research on RSs have a number of weaknesses [22, 141, 142]. RSs mitigation solutions can be incomplete as a result of research that narrowly focuses on one type of bias, such as popularity bias, without thoroughly examining all the possible sources of bias. The existing methods for detecting and correcting bias often rely on predefined fairness metrics or ad hoc adjustments [22, 72, 100, 141, 142]. There are, however, some limitations to these methods, including that they do not fully capture the numerous ways in which bias can be perceived in RS, resulting in solutions that can not be generalized across a variety of domains or be robust [143, 144]. Implementing fairness constraints in RS is still a developing area, with most proposed solutions being theoretical or limited to a specific use case. There remains a significant challenge in applying these constraints, particularly in dynamic and large-scale environments, that is often overlooked in the literature. Serendipity refers to the ability of a recommendation system to surprise a user with an

unexpectedly beneficial or interesting recommendation [145, 146]. Typically, it involves recommending items that users might not have actively searched for, but find valuable or enjoyable. Some literature highlights different types of serendipity; for instance, Boo et al [147]. have defined a new approach to overcoming the challenges of serendipity by effectively balancing exploration and exploitation, thereby improving the overall user experience. In recommender systems, balancing exploration and exploitation is essential since it ensures that users receive both familiar and unexpected suggestions by introducing new, unexpected items while still recommending familiar, preferred items. Serendipity of this type focuses on finding a balance between providing relevant and novel content that keeps the user engaged [141, 144, 147]. By incorporating diversity and novelty into recommendations, users are exposed to a broader range of items, thus preventing the formation of filter bubbles. Introducing users to diverse and unexpected content is an important aspect of this type of serendipity. Personalized serendipity can be achieved by tailoring unexpected recommendations to individual user profiles and preferences [148, 149]. The purpose of this type of serendipity is to align surprises with the tastes and interests of the user. A hybrid approach, which combines collaborative filtering, content-based filtering, and other methods, enhances the likelihood of discovering novel items, providing a well-rounded recommendation experience [150, 151]. This type of serendipity leverages multiple techniques to create opportunities for unexpected yet relevant discoveries. While it is acknowledged that improving serendipity is crucial, research frequently struggles to strike a balance between serendipity and relevance, lacks reliable assessment criteria, and struggles to successfully tailor recommendations based on serendipitous events. Furthermore, there is still much to learn about the intricate relationship between bias prevention and serendipity, as well as several implementation-related practical issues to resolve, particularly in large-scale, real-world situations. Further research on RS needs to be more thorough, relevant, and customized to address these limitations.

10 Conclusion

This literature review provides an examination of various strategies applied in recommender systems (RSs). Among the array of techniques, collaborative filtering (CF) methods emerge as particularly prominent. Notably, CF methods present advantages in terms of cost-efficiency and ease of implementation when compared to alternative approaches. It is noteworthy that contemporary trends in RSs have witnessed a surge in the adoption of personalized systems leveraging supervised machine learning methodologies.

These advanced techniques enable the customization of RSs to incorporate simulated individual user feedback, a practice instrumental in the generation of recommendations for new and pertinent informational items.

These strategies serve to foster serendipitous experiences within recommender systems (RSs) by incorporating novel information based on individual user input, thereby diverging from the reliance on biased collaborative filtering (CF) feedback, which emanates from user groups with shared preferences or comparable information items. Nevertheless, within the realm of RSs, challenges arise in the form of the cold-start problem, stemming from a paucity of user-generated ratings or data. This predicament necessitates the development of trust-aware RS techniques adept at mitigating biases inherent in RS recommendations, alleviating the risks associated with cold-start scenarios, and enhancing the probability of serendipitous content discovery. Consequently, this strategy finds widespread application in domains such as social media and various marketing RSs. However, it is worth noting that trust-aware methodology is not the exclusive approach employed in the domain of online RSs. Due to the proprietary nature of RSs maintained by commercial entities, there is a dearth of research publications disclosing the methodologies adopted in the routine operations of online RSs.

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Data availability This is a theoretical paper, and there are no data used for empirical study

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

Ethical approval This research is conducted in a manner that does not involve the participation of humans or animals.

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